Joint endpoint control and central cancellation in the recoil Barnes integral

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Abstract

An optimized left endpoint and the exact Gamma modulus identities control the negative vertical ray in a joint large-mass regime. Reflection and Stirling estimates control the positive tail relative to the sum of the two Bessel branch magnitudes. However, a central fixed-height window has an exponentially larger absolute norm than this envelope. Thus endpoint control does not complete the joint asymptotic theorem: the remaining estimate must preserve phase. These are self-reviewed analytic estimates and finite-point numerical checks, not a finite-G holographic dictionary.

1 Object, scale and remaining question

Set the de Sitter radius to one, $u = v = 0$, $\omega = i\alpha$, $K \geq 1$, $\alpha = Ky$, $0 \leq y \leq Y < \infty$, and $M = \nu^2/(2K)$. The fixed-mass $\kappa \to \infty$ limit has already been taken. This note concerns an internal chiral graph integral; no physical external state, operator algebra, insertion normalization or finite-G OTOC is assigned to it. Define

$$b = \tfrac{1}{2} - i(M - \nu), \qquad c = \tfrac{1}{2} - i(M + \nu),$$

$$Q_\nu(t) = e^{-i\pi t}\frac{\Gamma(t - i\nu/2)^4}{\Gamma(-i\nu/2)^4}\frac{\Gamma(b - t)\Gamma(c)}{\Gamma(b)\Gamma(c + t)}\Gamma(\tfrac{1}{2} - t + \alpha/2)\Gamma(\tfrac{1}{2} - t - \alpha/2), \tag{1}$$

$$U(t) = e^{-2\pi i t}(K/2)^{2t}\Gamma(\tfrac{1}{2} - t + \alpha/2)\Gamma(\tfrac{1}{2} - t - \alpha/2). \tag{2}$$

The contour runs upward from $-N - i\infty$ to $-N + i$, then right. Its integral includes $1/(2\pi i)$. Previously, the auxiliary integral was evaluated as

$$J = -K[e^{i\pi\alpha}K_\alpha(K) + i\pi I_\alpha(K)].$$

Use the positive scale

$$\mathcal{E} = K[K_\alpha(K) + \pi I_\alpha(K)], \qquad F(y) = \sqrt{1 + y^2} - y\,\mathrm{arsinh}\, y.$$

For negative α, evenness of the complete expression permits using $|\alpha|$ in this envelope. No estimate below divides by J. Its zeros are genuine obstructions to that error convention. Our hierarchy is $K \to \infty$, $\rho = K^3/\nu^2 \leq \rho_Y$, with a sufficiently small fixed positive ρ_Y. Statements about convergence of corrections would additionally need $\rho \to 0$; this note does not prove the full joint expansion.

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2 A multiplier estimate on growing compact regions

Fix $C < \infty$. On $|t| \le CK$, the quotient obtained by canceling the two small Gamma factors admits a continuous logarithm, with

$$\log(Q_\nu/U) = \frac{4i(t^2 - t)}{\nu} + O_C(\rho). \tag{3}$$

This is a growing-region estimate, not the direct use of a fixed-shift Gamma asymptotic. Here and below constants may depend on Y, C.

For completeness, apply sectorial Stirling to each large argument and Taylor expand its shift, using

$$\log\Gamma(z+w) - \log\Gamma(z) = w\log z + \frac{w(w-1)}{2z} + O\left(\frac{|w| + |w|^2 + |w|^3}{|z|^2}\right).$$

The segments $z + \theta w$ stay in a common closed sector away from the negative real axis, since $K/\nu \to 0$. The four numerator factors give $4i(t^2 - t)/\nu$. The leading logarithms combine to $2t\log(K/2) - 2\pi i t$; the remaining b, c terms are bounded by a constant times

$$\frac{K^2|t| + K|t|^2 + |t|^3 + K}{\nu^2}.$$

This proves (3), with the logarithm fixed continuously from zero shift after the small factors have been canceled. The first phase must be retained: K^2/ν need not tend to zero.

At a real $t = -N = O(K)$ this leading term is purely imaginary. At $t = x + i$ its real part is $-4(2x - 1)/\nu$. Consequently the left real base and the positive base satisfy

$$|Q_\nu(-N)| \le e^{C_Y\rho}|U(-N)|, \qquad |Q_\nu(x+i)| \le e^{C_Y\rho}|U(x+i)| \quad (x \ge \tfrac{1}{2},\ x = O_Y(K)).$$

On any fixed-height central strip, the modulus comparison is two-sided with factors $\exp[\pm C_Y(\rho + K/\nu)]$.

3 The optimized left base and its vertical ray

Choose

$$L = \frac{K}{2}\sqrt{1 + y^2}, \qquad N = \lceil L \rceil.$$

For sufficiently large K depending only on Y, $N > \alpha/2 + 1/2$, and both small Gamma real parts are bounded below by $c_Y K > 0$. Moving the left corner to this integer crosses no poles: the small-Gamma pole families start to its right, the numerator poles are at height $\nu/2 > 1$, and the $b - t$ poles have positive real part. The lower closure vanishes by the fixed-parameter vertical decay. Stirling on the positive real axis yields

$$|U(-N)| = 2\pi\exp[-KF(y) + O_Y(K^{-1})]. \tag{4}$$

Indeed, the leading exponent as a function of n is

$$g(n) = (n + \alpha/2)\log\frac{2(n + \alpha/2)}{K} + (n - \alpha/2)\log\frac{2(n - \alpha/2)}{K} - 2n.$$

At $n = L$, $g = -KF$, $g' = 0$, and $g'' = O_Y(K^{-1})$. Thus rounding to N changes the exponent by only $O_Y(K^{-1})$.

A lower bound for $\mathcal{E}$ valid even when $y \to 0$ follows directly from the standard positive integral [2]:

$$K_\alpha(K) = \int_0^\infty e^{-K\cosh u}\cosh(\alpha u)\,\mathrm{d}u.$$

Put $u_* = \operatorname{arsinh} y$. On $[u_*, u_* + K^{-1/2}]$, the exponent $-\cosh u + yu$ is at least $-F(y) - C_Y/K$. Using $\cosh(\alpha u) \ge e^{\alpha u}/2$ gives

$$\mathcal{E} \ge K K_\alpha(K) \ge c_Y \sqrt{K}\, e^{-KF(y)}, \qquad \frac{|Q_\nu(-N)|}{\mathcal{E}} \le C_Y K^{-1/2} e^{C_Y \rho}. \tag{5}$$

Proposition 1. *Let $B = M - \nu > 0$, $C_0 = M + \nu$, and $\lambda = 2\pi - N(B^{-1} + C_0^{-1})$. For every $s \ge 0$, with $N > \alpha/2 + 1/2$,*

$$|Q_\nu(-N - is)| \le |Q_\nu(-N)| e^{-\lambda s}. \tag{6}$$

In the hierarchy above, $\lambda \ge \pi$ eventually, uniformly in $y \in [0, Y]$, and

$$\frac{1}{\mathcal{E}} \int_{s_0}^{\infty} |Q_\nu(-N - is)| \, \mathrm{d}s \le C_Y K^{-1/2} e^{C_Y \rho - \pi s_0}.$$

Proof. Gamma recurrence and the standard moduli [1] give

$$|\Gamma(-N - iw)|^2 = \frac{\pi}{w \sinh(\pi w) \prod_{j=1}^{N} (j^2 + w^2)}, \quad w > 0,$$
$$|\Gamma(N + \tfrac{1}{2} + iw)|^2 = \frac{\pi \prod_{j=0}^{N-1} [(j + \frac{1}{2})^2 + w^2]}{\cosh(\pi w)}.$$

The reciprocal negative half-integer formula follows by recurrence. In the ratio to the base $s = 0$, the explicit exponential gives $e^{-\pi s}$. The four negative-integer Gamma factors give at most $e^{-2\pi s}$, since their rational factors decrease and $\sinh(a + b) \ge e^b \sinh a$ for $a, b > 0$. The two large half-integer hyperbolic factors contribute at most $e^{\pi s}$, because $\log\cosh$ has derivative bounded by one.

Each polynomial ratio is bounded using $\sqrt{a^2 + (B - s)^2} \le (1 + s/B)\sqrt{a^2 + B^2}$, and likewise with $C_0 + s$. Their combined contribution is at most

$$(1 + s/B)^N (1 + s/C_0)^N \le e^{Ns(B^{-1} + C_0^{-1})}.$$

The remaining small Gamma ratios are at most one, by $|\Gamma(a + is)| \le \Gamma(a)$ for $a > 0$, a consequence of Euler's integral and the triangle inequality. This proves (6). Finally $N(B^{-1} + C_0^{-1}) = O_Y(K^2/\nu^2)$, and integration gives the stated bound. □

The short connector from $-N$ to $-N + i$ has the same $O_Y(K^{-1/2})$ envelope bound, with factor $e^{C_Y(\rho + K/\nu)}$: use $|\Gamma(a - i\tau)| \le \Gamma(a)$, $0 \le \tau \le 1$, and $|e^{-2\pi i(-N + i\tau)}| \le e^{2\pi}$. These statements control the vertical pieces only. The negative horizontal part of the central segment is not included.

4 The positive base and the entire positive tail

Set $c_Y = 8 + 2Y$, $R = c_Y K$. Reflect both small Gamma factors:

$$U(t) = \frac{\pi^2 e^{-2\pi i t} (K/2)^{2t}}{\cos \pi(t - \alpha/2) \cos \pi(t + \alpha/2) \Gamma(\frac{1}{2} + t - \alpha/2) \Gamma(\frac{1}{2} + t + \alpha/2)}.$$

On $t = x + i$, the cosines are bounded away from zero. For $x \in [R, R + 1]$, let $r = x/K$. Uniform Stirling gives

$$|U(x + i)| \le C_Y e^{-K h_y(r)}, \qquad h_y(r) = (r + y/2) \log[2(r + y/2)] + (r - y/2) \log[2(r - y/2)] - 2r.$$

The function is even and convex in y on $|y| < 2r$, hence $h_y(r) \geq h_0(r) = 2r[\log(2r) - 1]$. Since h_0 increases for $r \geq c_Y$, the true supremum obeys

$$H_R := \sup_{x \in [R, R+1]} |Q_\nu(x+i)| \leq C_Y e^{C_Y \rho - K h_0(c_Y)}.$$

Use the previously established exact-recurrence tail lemma: for $D_Y = \max(2, Y^2/2)$ and $\nu^2 \geq 32(D_Y + 1)K^3$,

$$\int_R^\infty |Q_\nu(x+i)| \, \mathrm{d}x \leq 10(R+1)H_R.$$

Together with (5) and $F(y) \leq 1$, it follows that

$$\frac{1}{\mathcal{E}} \int_R^\infty |Q_\nu(x+i)| \, \mathrm{d}x \leq C_Y \sqrt{K} \exp\{-[h_0(c_Y) - 1]K + C_Y \rho\}. \tag{7}$$

The positive endpoint is therefore exponentially small in this normalization. Sampling H_R numerically is not used as proof of its supremum bound.

5 An exponential obstruction to central absolute bounds

Fix $y > 1$. For $t = Kx + i$, $-y/2 < x < y/2$, reflect just the Gamma factor with negative real argument. Uniformly on compact subintervals of this interval, positive-real Stirling then gives

$$|U(Kx+i)| \asymp_y e^{K A_y(x)}, \qquad A_y(x) = u[\log(2u) - 1] - v[\log(2v) - 1], \quad u = y/2 - x, \quad v = y/2 + x.$$

The periodic reflection prefactor is bounded above and below by positive constants at height one; it cannot be discarded in the complex integral. Direct differentiation gives

$$A_y' = -\log(y^2 - 4x^2), \qquad x_* = -\tfrac{1}{2}\sqrt{y^2 - 1}, \qquad A_y''(x_*) = -4\sqrt{y^2 - 1}.$$

At this maximum,

$$A_y(x_*) = B(y) := y \operatorname{arcosh} y - \sqrt{y^2 - 1}.$$

For any fixed small $a > 0$, the window $W_K = \{t = q + i : |q - K x_*| \leq a\sqrt{K}\}$ therefore satisfies

$$\int_{W_K} |U(t)| \, |\mathrm{d}t| \geq c_{y,a} \sqrt{K} \, e^{K B(y)}.$$

Let $y_c = 1.5088795615\ldots$ be the zero of F. For fixed $y \geq y_c$, the standard large-order Bessel formulas [3] give $\mathcal{E} \leq C_y \sqrt{K} e^{-K F(y)}$. Thus

$$\frac{\int_{W_K} |U| \, |\mathrm{d}t|}{\mathcal{E}} \geq c_{y,a} e^{K \Delta(y)}, \qquad \Delta(y) = B(y) + F(y) > 0. \tag{8}$$

Indeed $\Delta' = \operatorname{arcosh} y - \operatorname{arsinh} y < 0$ and $\Delta(y) \to 0$ at infinity (its leading term is $1/(2y)$). This proves positivity on $y > 1$, in particular on $y \geq y_c$. The two-sided multiplier estimate transfers (8) to Q_ν, at cost $e^{-C_Y(\rho + K/\nu)}$.

This does not refute a joint signed-amplitude asymptotic. It shows that the fixed-height absolute norm can be exponentially ill-conditioned relative even to the positive branch envelope. Multiplying a pointwise polynomial remainder bound by this norm does not yield the desired polynomial envelope error. The same obstruction applies to the retained quadratic phase: its modulus on this strip changes by only $e^{O_Y(K/\nu)}$.

6 Numerical checks and continuation

All data use the exact inherited kernel and $\nu = 32K^{3/2}$, so $\rho = 1/1024$ is fixed. This tests bounds in a joint hierarchy, not convergence as $\rho \to 0$. For $K = 8, 32, 128, 512$ and $y = 0, 1, 2$, we check the real-product vertical identity against complex log-Gamma evaluation at ten values of s, including $M - \nu, M, M + \nu, 2M$. We also check the reflection identity on three positive base points per parameter pair. These are finite-point checks; the proofs above provide the continuum estimates.

The central computation integrates absolute values only, over $|q + \sqrt{3}K/2| \le \sqrt{K}/4$, at height one. Unit-or-shorter cells use 20/32 Gauss–Legendre nodes and 55/75 decimal digits; the last row is also checked at 95 digits. The largest relative order change is below $7.4\,10^{-27}$. No signed full integral or cancellation-resolved relative error is computed in this experiment.

K	$\int_{W_K} \lvert U\rvert/\mathcal{E}$	$\int_{W_K} \lvert Q_\nu\rvert/\mathcal{E}$	$K^{-1}\log(\int_{W_K} \lvert U\rvert/\mathcal{E})$
8	$9.33072\,10^{2}$	$1.00797\,10^{3}$	0.854810
16	$5.65331\,10^{3}$	$5.94872\,10^{3}$	0.540000
32	$2.77261\,10^{5}$	$2.86796\,10^{5}$	0.391647
64	$7.95113\,10^{8}$	$8.12952\,10^{8}$	0.320219
128	$7.15645\,10^{15}$	$7.25825\,10^{15}$	0.285209

The analytic rate is $\Delta(2) = 0.2506620134229251\ldots$; the finite-$K$ rates include large constant-prefactor effects. The 300 checks consist of 120 vertical identities, 120 vertical inequalities, 12 decay-rate checks, 36 reflection identities, 10 order refinements and two precision refinements. Base and local-multiplier samples are recorded separately, without invented universal numerical constants.

The next task is a phase-preserving estimate of the central integral, for example by a justified analytic branch decomposition or contour deformation with explicit pole bookkeeping. Only after a complete two-branch joint estimate should one restore u, v, transverse tails and the physical external interfaces. Finite G, state identification and a microscopic holographic dual remain open. No priority claim is made.

References

[1] NIST DLMF, Sec. 5.4, Sec. 5.5, Sec. 5.11: Gamma moduli, recurrence, reflection and sectorial Stirling.

[2] NIST DLMF, Eq. 10.32.9: the positive real integral for K_α.

[3] NIST DLMF, Eqs. 10.41.3–4: large-order modified-Bessel expansions, used here only at fixed $y > 0$.

Reproducible files:
`code/recoil_bessel_endpoints.py`
`results/stage4/recoil-bessel-endpoints/diagnostics.json`
Previous analytic input:
`notes/stage4-recoil-bessel-cancellation.md`